

Meeting Transcription

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Meeting participants: ADITYA ZINE, Deepak Goyal, Deshal, Dharmender Singh, Gopinath M A, Hemant Rade, Naveen Ta, Niharika Modi, Pati Mahi, Raghunath Meher, Ritesh Ramesh, Rupali Bandgar, SHOVAN PANDA, Savan Kardani, Srikanth Rajapuram, Sudhakar Davane, Surendra Jami, Triveni B, VEMULA MAHIPAL, Vaibhav Awasare, Venmathi Ravi, anuja bhanarkar, anurag ojha, sanat sharma, urmi patel, vivekanand dhumaskar

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Highlights

- 19:37 Deepak Goyal: Spark notebook, which will take the data from this customer folder. And convert the CSV into a parquet file format and save it into over bronze layer. How I can do it. I click on this here. There is a developed tab. Click on this develop tab. Click on this plus sign here and here you see notebooks.
- 01:36:17 Deepak Goyal: address ID Okay, I think so. We can combine base of the customer ID. I think so. Okay. For this customer because address and let's just see what is there in the customer table as well? Just run that too as well. So customer address and customer does not have any address ID column so we can join the data frame 1 but join era Frame 2 on In summary, tell me the syntax. I think I'm forgetting something.
- 01:41:59 Deepak Goyal: Okay, so you can see that now we rename these two column because these two column coming. So we just rename with modified data address GUI id address and then we join it on the customer ID column and we save the data into the gold layer. So if we come down here onto our goal layer. Landing folder project 4 I should see a gold layer I can see gold layer.

- 01:48:51 Deepak Goyal: When I run this bronze schema, it has created a one bronze schema. Let me create similarly a silver schema. Echo this will get the silver schema. And let me create a one more schema. I call it like a goal
- 02:13:41 Deepak Goyal: Say connect and then whatever be it is asking to connect there. You just put the synapse. Server name? Okay. So this this is how you connect and say okay and they will ask you the credential you choose the Microsoft account and whatever be the current account you just connect through that. So automatically you will have an access to see all that data. And once the data is there you select which table you want to see come down select that table select the chart which you want to print or you know create and that and then your power will be I think get connected. Thank you so much. I know we have extended little bit.
- 02:14:18 Deepak Goyal: But this something we need to finish up. Okay for the next week onwards. Will still have a session but next week my new batch is getting started. So we probably will not able to do it on the same time. We push the I feel at least two weeks thing left maximum so we cover these two week thing on to Coming week in the next week, but we're doing the sessions into the evening time.
- 02:24:11 Deepak Goyal: A logic is simple whether it is a POC or it is not a posit doesn't matter. Okay, it's important that what you're trying to do.
- 02:26:11 Gopinath M A: Deepak assignment solution 7 is missing in the portal.

Transcript

00:00 Deepak G.: Okay. So let's go to our Azure synapse we try to create some kind of a layer silver layer within this little bit layers like a go on bronze synapse world and we try to pull the Then we're going to use the the synapse. data using the ADF P's of the And then finally, we'll see the power external tables of the synapse. synapse with the power bi and pull bi connect.

00:06 Deepak G.: We'll see that how we can connect the all the visualization. the data into the power bi to create today to have this project. entire integration we can add remove So this is the agenda for mainly for It's more I I should not call it a

requirement around it. project but at least could show that more of you know, business Studio. Now if we imagine that we have an on-premises database and the on-premises database is our same database that we have used so far doing this entire course, so we use that same SQL Azure SQL DB for now to pull the rate out of it.

00:24 Deepak G.: Okay, and then we're going to push it into a 180 list location. via synapse itself So let it come here. Okay. So this is our synapse. we come down to our integrate tab because what I want to do is I want to connect to my Azure SQL TB okay. I assume that this is my on-premises database and I'm first trying to pull the data from on-prem to the cloud.

01:02 Deepak G.: Okay, if we exactly do it for on-prem to the cloud we should have created a self-hosted integration runtime and pull it through that but just for the sake of you know. In this sake of time because otherwise we'll spend a lot of time in creating that self first indication on time. I'm just using the Azure SQL DB only. Okay. So let's give the pipeline name. I call it like injection.

01:31 Deepak G.: SQL DB to ADLs and let's say I'm doing it in CSU format itself. Okay. Now coming back to here. I use again a copy activity. So this is my copy activity. Now this copy activity. What I'm trying to do is in this copy activity. I want to pull the data. so equal DB To ADLs. That is what I'm trying, okay. I need a source. I don't have any data set.

02:12 Deepak G.: So I click on this plus new to create a data set. I use SQL DB. So this is my Azure SQL database. I select that again. This is a pretty simple thing which we did many many times in our ADF. Equal to be data set. It needs the link service. We don't have any link service. So let's create us SQL DB link Services as well. Right. Now we are connecting to the Azure Cloud only so that's why we are using the auto resolve integration runtime.

02:52 Deepak G.: The let's use this server name server name something Mission batch 7 DB server. Which is our default. Database was mission ad 7 DB username. If I remember it correctly, I don't know. Change the password. and remember A change my usual password. Okay. Let's come down here and quickly go to that database server. To reset that password. So this is also a way if sometimes you forget the password. You go down to your database.

04:04 Deepak G.: Go to So this is my database server. And here on the top of it

here. You see that reset password link click on the preset password link. I forget my original password.

04:37 Deepak G.: s*** save so our password get saved. Now what I do is I come down here again. And this time I give the password which I have recently. set and then if I see the test collection, ideally they should come out to be successful. All right, and let's say create. Now this will create our link service which will connect to the Azure SQL DB. Remember one thing we are doing all of these things not into the ADF world.

05:22 Deepak G.: We are doing this into this synapse world, okay. I said this order table for now order details. It's okay. Let's preview the rate of this order tables. Okay. So this is our order table data looks like I keep it here. let's try to save this information. So I'm just coming here into the sink where I want to save you want to save into our blob storage location.

06:28 Deepak G.: So I go down to this blob story this time rather than saving in terms of a CSP or anything. Okay. Let's see. Let's save it in terms of Is itself so I keep it CSV here. Okay, I say continue. I give ADLS. data set and I'm just using this default connection link because remember when we create the synapse workspace account with the synapse workspace account you get a one default.

07:09 Deepak G.: Ad less account get attached that is where we seeing all the landing folder and all so we am using we are going to use the same thing only. That default one. And under the default one. Let's say I have this Landing folder. Okay. I have a lot of other folder. Let's say okay for now. Say continue. And give the name of this ADLS. data set in services this Folder is can Landing. Say, okay. First row is an header. And let's say okay for now. Okay. Okay.

07:53 Deepak G.: I'll just quickly tell you what we have done so far we have. Get into this synapse world. So this is synapse under that synapse. I go down to this integrate tab under that integrate tab. I click on plus here. I created pipeline once I created a pipeline I search here in the activities copy activity. I drag out the copy activity. Okay, once I have this copy activity. I have to create the source.

08:19 Deepak G.: So I pull in the source as my sequel database and in that I'm right now here. If I see the SQL data set. Let's make it parameterized so that we can later on change as well. So I give it like Emma name server schema is always cell t i given another. Parameter that is stable name. In the connection instead of

putting the details will enter manually. We add Dynamic content.

09:00 Deepak G.: So this will come from schema name coming from schema name variable and say the table name is also coming from Our this table name? Okay. So if we little bit minimize the screen this looks something like this. Okay, so our SQL database data set we have created and we have parameterized it. We're giving this schema name and table name at the runtime. So we keep it here. Somebody has a quick question. Generally.

09:42 Deepak G.: I don't allow that but yeah, you can put on the question under the chat. I'll I'll see. Okay, actually. Okay, and here if we see. This data set I should see both of these two things schema name and table. Let's keep our customer table itself for now. Okay. I just keep my customer table itself. And in the sink sync Also, let's made it parameterize. I wanted to save it at a but specific location. So I just add the parameters.

10:20 Deepak G.: I gave the folder name as the parameter. Okay, so this is my folder name I go down to this connection. now here, I just pick out the folder name and I said okay that now if I go to my pipeline here when I'm using this data set it is asking me a folder name where I want to save it. So folder name, let's create what? We create create.

10:57 Deepak G.: object 4 slash raw layer or let's project fold Landing folder. Okay ending slash customer. This will go inside the project folder and under the project folder. We have a landing folder. And in that we have a customer folder there our data should go got it. So let's say if I come down here. Now and if I execute this, so if my pipeline get executed successfully. My data will move from SQL database.

11:35 Deepak G.: Into a one folder location that is project for landing customer. Okay. So this is my customer pipeline. Let it get run it is still running. Remember we are doing this not into the not into the ADF world. We are doing this into the synapse word. Okay, so you can see that our pipeline is running let it get run. Meanwhile, it's run successfully. You don't forget to click on publish all to save all your changes.

12:06 Deepak G.: So we click on publish so changes get see so we have added the customer. Let's add a one more table. We call it customer address table. Okay, so we'll do the another thing. We just add let me just quickly clone this pipeline only to make the things faster. So I clone this pipeline. So everything will

get attached. And lastly I call it SQL DB a deal is to CSV.

12:34 Deepak G.: customer address this is for customer address now in this I will make a one small change. The table name I change from customer. to customer address and sync location also the folder name change from customer to customer address, right? Oh, this will also going to be very fast work now if I just you know, debug it. This will going to pull the data from the customer address table and push it into a folder location call customer address because our database Remains the Same our sink main container.

13:18 Deepak G.: Also Remains the Same you just see in the folder name and that is the advantage of making our data sets parameterize. So now you can see that making a second pipeline was we easier for us. Okay. So this is also running and this is also get successful so far we created two pipelines the work is that will move the data from our on premises. Example was I mean we are here using a cloud database but imagine that it is pulling from on-premises databases.

13:50 Deepak G.: So now we pull the radar for using these synapse pipelines from the from your database servers into the cloud location. That is ADLS location. Now as we are into the synapse world, we can Traverse everything from here itself. How now this was so far was we are in into the integrate tab if I go down to this data tab So under this data tab, I go down to this linked here.

14:19 Deepak G.: And if I open up my this storage account. and if I see my Landing container There I should start seeing the details. So in this Landing container. You see so this is my Landing container under that Landing container. I can see now my folder that is project for go down to that project folder location. Under that. We have a landing folder in Project 4 and there we got a customer folder.

14:54 Deepak G.: In the customer, we got customer data. in the Customer address folder we got the customer address file got it so our input data are available. Got it. So imagine that the ADF pipeline has done the work of moving the data. Got it. Now, let's just Create a one. Spark one spark notebook which will convert which will read this Landing there. Whatever the data is coming or drop by this.

15:37 Deepak G.: our ADF Pipeline and it will convert it into a parquet format and put it into our Role folder layer okay, or these we call it like a bronze folder

location? Okay. So let me just at least draw a diagram and show you what exactly I'm trying to do.

16:25 Deepak G.: Whoever has a question they can put down into the chat. Okay, because we are doing quite practical. So if somebody has a challenge here itself they can ask. In the chat. Okay. So what I did. We have the database. Now we use. our synapse ADF part that is pipeline part to pull the data from database and push it into a one folder location into an ADLS account.

17:05 Deepak G.: Now what is the next step I'm trying to do is we got a folder here into a folder name called. Landing and our data is in terms of CSV file Okay. Now what I'm trying to do is I convert this landing CSV into an another folder. into the Adidas location itself, and I want to call this folder as my bronze layer generally in the bronze layer. We don't do any you know. We don't do any. Transformation or anything. We just want to lend up as it is. So I want to use here the spark.

17:47 Deepak G.: notebook to just convert my CSV into the parquet file format and save it into a bronze layer folder then later on. What I want is after our bronze layer will get create I want. another spark notebook Which will do some kind of a transformation or clean up. and whatever be the result comes out out of the cleanup put it again into an idealist folder location, which we call as like a silver layer Right all this we are doing this Transformations using the spark notebooks.

18:30 Deepak G.: Okay, and later on what we're going to do is Because we are using. These spark sorry we are using the synapse. So what I'm going to do is I will also correspondingly create an external table for bronze layer an external table for the silver layer. This is we are going to create for the data analyst perspective so that data analyst can you know data Engineers scheme, they pull the data then they use the spark to create the bronze layer silver layer gold layer, then the returnless come and they create the bronze layer tables silverware tables and do the you know, Analytics surrounded if they want to do it.

19:08 Deepak G.: Okay, so we'll see in the both the way through the spark when then through the our external table way, okay. So let's just come down here. So this folder got created everything looks fine so far. So I close all other tabs, which is not needed. Now. What I want is I want to write a one.

19:37 Deepak G.: Spark notebook, which will take the data from this customer

folder. And convert the CSV into a parquet file format and save it into over bronze layer. How I can do it. I click on this here. There is a developed tab. Click on this develop tab. Click on this plus sign here and here you see notebooks.

19:58 Deepak G.: So click on the Node book because we want to write as Park code. Okay. So this is a spark code now. the beauty is that sometimes

20:26 urmi p.: frozen

20:29 anurag o.: we lost you.

20:40 sanat s.: Got stuck

20:52 Deepak G.: one min

20:55 Rupali B.: no sound

20:58 Srikanth R.: lost you

21:00 Naveen T.: Lost call

21:14 Deepak G.: one min guys chekcing

23:44 Deepak G.: memorable now

23:46 Deshal : yes

23:51 anuja b.: yes

23:52 Venmathi R.: yes

23:52 Srikanth R.: yes

23:52 sanat s.: Yes

23:54 Deepak G.: Thank you.

23:55 Surendra J.: Yes

23:58 SHOVAN P.: Yes

24:06 Deepak G.: Okay, I'm really sorry for this hiccup. I hope my okay. My screen is not visible yet. Seems like come on. Okay. Is I hope it is visible now. the fortunate stuff is that we are saved everything so In this connection hiccup. It doesn't last all our data. Okay, so coming back. Let this screen get loaded up. Meanwhile, we were on to the screen. We are pulling the data from ad database and putting into the ad list location.

25:07 Deepak G.: Now, the next thing what I'm trying to do now is I'm going to create a notebook and in this notebook. I'm not doing any big thing because I want to create my bronze layer in the bronze layer. I just don't want to you know, do any manipulation whatever the time getting it. I'm just dumping as it is into my bronze layer. Okay. So in the our this part notebook, I'm not going to do any big

thing.

25:31 Deepak G.: I'm just right now one simple logic which will read the CSV file converted into the park. Okay and save it into the grounds layer, but when we do bronze to the silver, we'll do some kind of a transformation and then only we'll save it. So first notebook will be our bronze layer. Notebook that will be going to be a very simple notebook where we are not doing any big change.

25:56 Deepak G.: here and there synapse is taking too much time to load. Let's see. how

26:30 Deepak G.: let's just load it. Taking too much time in loading. No, no what happened to that synapse? Let me try to open it from here

27:07 Dharmender S.: your network seems to be slow i guess

27:09 Deepak G.: again. Maybe that will solve the problem. It's not loading up at all. Let's see here.

27:25 Dharmender S.: voice is coming and going too

27:30 Deepak G.: If entire portal is getting downward. See, this is also getting very very slow. click on here open Snap Studio this one not yet loaded. This one not yet loaded. Let's connect to this. Let's say hopefully this will get loaded up.

28:16 Deepak G.: Somebody is saying that there is a problem in my voice. So please confirm is audio coming fine. I saw the chat message that there is a some voice problem, please confirm.

28:28 Venmathi R.: yes

28:36 Rupali B.: yes

28:37 SHO VAN P.: Yes

28:40 Gopinath M.A.: yes sometimes audio is fluctuation

28:42 Deepak G.: You are saying is a problem or it's all okay. always fluctuation is it because that I am moving I'm mouth here and there too much Okay. Just please confirm is all good now or not. Good yet. Please confirm our synapse also

29:12 Venmathi R.: all good

29:12 Deepak G.: loaded up.

29:14 Savan K.: good

29:15 Niharika M.: Good

29:15 Deepak G.: All good, everyone. Okay.

29:18 sanat s.: Allset

29:19 Deepak G.: Back to the business.

29:19 Raghunath M.: yes good

29:20 Deepak G.: Okay. So this was our pipeline. Let's go down to our Pipeline and see

29:23 ADITYA Z.: All good

29:24 Deepak G.: our pipeline. So we have a 2 pipeline one was pulling the data for our customer table. Another pipeline was pulling the data for our customer address one. Okay, and then we get down to our data tab when we go down to our data type. Click on the link here go down to this storage account open that storage account go to The Landing folder another Landing folder. We have to look for a folder. The folder name was.

29:50 Deepak G.: Project 4 so we go down to that project fold folder location here is Project 4. Go down to that. We saw Landing folder and there we saw two folders customer address customer contains the customer data and customer address folder contain the customer address data. Okay, so now let's do one thing. we have to create a one notebook where we can read this data inside this folder. So there is a one option.

30:17 Deepak G.: What I can do is I can right click on this folder here itself. And I can see okay. The option is not coming. Ideally this should come but anyways. Let me rename this file for a timing to see SV. Tells LT customer dot CSV. Now see the moment. My file name is dot CSV. I right click on this. I can see an option for the notebook as well.

31:00 Deepak G.: So what I'll do is we can make a one tweak here as well because it seems like we want to have and do the CSV format. So what we should do is in our pipeline itself. Let's make a one small change here in the pipeline we go down to our sync location. we open this data set and we add here file name parameter as well. So I'll add up one more parameter. I call it like file name, okay.

31:32 Deepak G.: And here under that file name, let's choose that file name parameter. so that we should get the dot CSV name because by default it is coming as a DOT txt. So rather than it should come as dot txt. Let's have a DOT CSV itself proper one. So I come down to now my this pipeline pipeline. I'm all automatically getting that. So I call it a name called customer dot CSV. Let's read on the pipeline.

31:58 Deepak G.: So now it will create a file with the customer dot CSV name correctly? Okay. Secondly I go down pipeline in this pipeline that is customer address one. They are also I'll set the file name as customer address dot CSV and run this pipeline, too. So now we're both. The pipeline is executing the pipeline one got successfully executed. That is pulled the data into a folder location. You can see that it is copy 847 records.

32:32 Deepak G.: in the folder location done Similarly we go down here onto our second pipeline. And here also you can see that this is also copied the data this time. The second pipeline is pulled four one seven records. Okay, and if we go down to this Landing folder location here now if I just go to this Landing folder go to customer folder. I can see two file customers CSV file and this sales LT1.

33:03 Deepak G.: So for now, let's just delete sales 81 because we don't need it anymore. So delete that let's have just on the customer CSV here. And if you go down to customer address. Let's have a just a customer address CSV here. So let's delete that also the txt one got it. So now we are landing folder and in the landing folder. We have a two entities customer and customer address and both of them contain its own CSU files.

33:35 Deepak G.: So now right click on this and say Notebook. We right click on that CSV file right click here and say notebook. So you are getting an option load to data frame and automatically, it's create a 1 spark notebook now for running in spark notebook. We need to have the cluster attached to it. Right so we have added the on the top yesterday. We created one clusters spark cluster. I attach that cluster and here.

34:13 Deepak G.: You can see that a code has been written automatically. What is the code? Is there which has been written sparked or Red Dot load and this is the folder location from there. It is reading the file. Format it is saying CSV. And we have a header equal to True also, so I just set. here header equal to 2 comma in first schema also equal to true so I'll set the inverse schema header and everything equal to true.

34:45 Deepak G.: So what data frame get create we can just also move this bracket to here also, so this is our entire line. The path is coming very big. That's why you are seeing it. If just for example, I remove the path you can see that the code is this one only sparked or read dot load then your path name for Matt CSV

header equal to true in first schema is equal to true. So I just please the path again.

35:13 Deepak G.: And then we just write a simple logic to you know, write it. So I just say DF dot, right? Dot option when we say safe path is equal to now we have to give the path. So what part I have I will give I'm giving a one in the project folder location. I'm giving a One path. I'm calling that path as a Bronze layer, so I create a folder called bronze layer and inside that I create a customer folder there. I want to save the file now. I have not given any format.

36:01 Venmathi R.:]]\

36:04 Deepak G.: So whenever you don't give any format by default is a parquet file format. So this will going to read the CSV file and save that CSV file into my customer folder location. Bronze layer and in the bronze layer, it's a customer folder location. They are the data gets saved and this will going to save the data for in the park a file format. So this is the code one. I'll add a same piece of a code.

36:32 Deepak G.: Because this will copy the data for customer table. And we have to do the same thing for the customer address. So here I just change the name here. customer address Landing customer address folder And the customer at results CSC file, it will read and save it into. Customer address folder so I'm not done anything big first. I reading the customer financing into the parking format below. I am reading the customer address file and saving it into the customer address folder location that too into the bronze layer.

37:13 Deepak G.: Okay now I don't want to execute this. Notebook from here. Let's okay. Let's let's do one thing. Let's remove this custom address thing. Let's have this notebook only for customer for now. Okay, we call that custom notebook here. as the customer Spark Notebook, so you remember that? Okay, so we give the name customer notebook only now. What I will do is I go down to my pipeline here itself.

37:58 Deepak G.: And what I can do is either I can create a separate pipeline. integrate we have to do the things automatically, right? In the databricks world when you create the notebook you schedule the notebook, right? But here you don't schedule The Notebook. What you do is you create the notebook and you create a pipeline and you call the notebook from the pipeline that also makes

sense, right? That is what we're going to do here. So I click on the pipeline.

38:31 Deepak G.: I call this pipeline as bronze layer summer now in this pipeline I can have us a one activity which activity you don't find it into the ADF world. So in the Ada, you don't have the synapse activity. So I'm using that synapse activity. Okay here what I can do is I can call the synapse node book. So I go down to the setting tab. And here you can see that I automatically getting an option in the bottom to call the notebook and here you can see that we have created a one spark notebook. For the customer.

39:13 Deepak G.: So this is options automatically coming in. I choose this option. Customer spark load book now, it will ask me to choose the spark pool in which it's going to call. So you already created a one cluster yesterday spark cluster. So I choose that same spark cluster. I keep it. The size is small only. Okay. And then we keep it. Here it is. Okay. So now if I execute this pipeline.

39:47 Deepak G.: Now this pipeline ideally should do what this is going to call a spark notebook, which is this part notebook The Notebook which we have created The Notebook looks like this. So this notebook, which is just reading the data from the CSV and copying it into the bronze layer. Now all of these things we are doing into the synapse. So let it get execute again. I'll repeat everything so that everybody is come into the sink.

40:14 Deepak G.: So don't worry if you feel that you miss something I run little fast. Don't worry. So here you can see that. This is The Notebook which is running. and you can see the execution as well the notebook execution the way you see the spark execution right spark database execution. Similarly here. You can also see that as well. It's still loading up. So let's go back. Several spark notebook is running once our spark notebook get executed. We'll see the output.

41:01 Deepak G.: So I'll just summarize what we have done so far. Okay, so that everybody comes into the thing see I trying to solve this problem where I'm pulling the data from database pushing it into a one folder location into the cloud world. I use a synapse pipelining. So how I use synapse pipeline. I go down to my CSV. I this pipeline I created up first pipeline in this pipeline.

41:25 Deepak G.: I created a one source source was a normal hour. SQL DB data source and in that I created two parameters scheme management table name. So here I'm giving that schema MSL T and table name is customer so that it can pull

the customer data in the sync location. I just give this folder location project for landing customer folder and purposefully, I want to read CSE file and synapse expect that it automatically give you the option to write, you know code for you.

42:01 Deepak G.: So it is explicitly expect that it should be of a CSV format. So I just give the final name as well customer dot CSV. So this will pull the data for the customer. Similarly. I created another pipeline because I have another table to bring in into the idealist location. So I created another pipeline that will pull the data for sales LTE customer address table. And the customer address data, we see it into a sync location.

42:29 Deepak G.: That is Project full Landing folder customer address folder inside it and the file name customer address.csp. So now we got the input available into our Landing. Okay. Now we create a three layer architecture one is bronze layers silver layer gold layer. So in the bronze layer We don't do any manipulation, right? So whatever we are getting into our this Landing folders and just copying as it is into our bronze layer.

42:57 Deepak G.: But in the our Landing folder, it is come coming in the format of the CSV. We want to convert it into the format of the Json. Sorry a parking. So I created instead of creating doing through this. Pipeline ID through the spark code and however, I write it I come down here onto the data tab. I go down to this Landing folder location. Under The Landing folder location I go down to that project 4 landing and customer.

43:30 Deepak G.: I right click on the customer and there I automatically get an option new notebook. Click on load to data frame the moment. I do this. I got this notebook written something like this. So it gives me an entire path and the first line get return where I set it format equal to CSV header equal to true and in first schema also equal to true and then what I did is I don't have to do any manipulation for the bronze layer.

43:55 Deepak G.: So whatever I read it from this Landing folder location, I write it as it is. I say data frame dot right dot save and this is the location. I created one branch layer folder location, and I'm trying the customer return to this bronze layer folder location. So that's how we have converted our bronze layer thing. Okay. Now who will execute this code. So rather than executing this manually, I have scheduled this code how I schedule this spark code. I go down to this

integrate tab. I created a one more pipeline.

44:33 Deepak G.: I call it bronze layer pipeline under that bronze layer pipeline here. There is a one activity called synapse. It was not there in the EDF world. It is here because it's a synapse. So I use this notebook activity now in this notebook activity. I can call the synapse spark notebooks. So automatically I call that customers Park notebook, which we have just created pause the cluster information and then I click on execute it so that it should get execute.

45:03 Deepak G.: Now, you can see that this notebook also executed successfully and if everything got successfully let's just see output what the output looks like. So it is saying that this has been executed successfully there is no output to print because I was just right saving this data into a bronze layer customer folder. Okay. So what I should do is I go down to my data tab again And if I see that if everything goes fine, if I go to The Landing folder container.

45:37 Deepak G.: Under that if I go to the project for folder location. Under the project 4 I can see that bronze layer folder got create under that customer folder should got create. And under that a file should get saved in the form of parquet file format. So I go down to the customer and I can see that. A file gets saved into my raw layer. I mean bronze layer that is of the parque file format got it.

46:04 Deepak G.: you can see that we are using nothing So our customer data gets saved and but entirely the synapse part we are using a we are loading the data by ADF, but we are using the synapse integrate pipelines. Then we are creating the notebooks for that. Also. We are using the synapse and then we are you know. Scheduling that notebooks also through the synapse also. What I can do is later on if I needed to you know, right now we have not scheduled it. We've just triggered it.

46:34 Deepak G.: But if I want to run it into some triggered manner this notebook. Also I have to just click here at trigger and I can set the trigger that this notebook will run every hour every once in a day or something like that. Okay, so that is fairly possible. Got it. Now same thing. We're going to do it for the hour another notebook as well. That is customer address notebook how we going to do it.

46:59 Deepak G.: We go down to this Landing folder again. We go down to four project four folder location go down to Landing go down to customer address and there we see customer address CSV. I right click on this. Say notebook load to

data frame. So automatically a code bill has been written for me, which is loading the customer address data. So I just remove this line because I want to set my header equal to true so I set my header equal to true.

47:30 Deepak G.: comma I say infers schema also equal to true and and I say bracket here. I close that and I remove this extra packet from here now. I don't want to display this. I want to load this whatever the data we are reading also into the bronze layer. I'll say DF dot right? Dot save and under save I can see my path is equal to so whatever be this path is I copy this entire path.

48:07 Deepak G.: Now we are not giving the CSV path here. We're saving into the parquet file format. so if you can check it the path, which I'm going to give is project for the folder name is bronze layer under the bronze layer folder. I could customer address folder. And file level come automatically so I just save this much. Okay. So what I did read the reader frame from our Landing folder and writing it to the bronze layer.

48:40 Deepak G.: Dataframe.right.save, and I save it into project 4 bronze layer customer address folder location. Just change the name of this pipeline so notebook. So I call it like customer address notebook. Okay, so this will become my customer address notebook. I save this I come down here. I create a one more now click on integrate tab. click on one more pipeline Add the resource create a pipeline. and this pipeline is going to Again uses synapse notebook and this time this will pull the data for what I call it.

49:25 Deepak G.: bronze layer customer address pipeline. So this will call the customer address notebook under the notebook go to the setting tab. Select the notebook name. I select the customer address notebook. I select the spark cluster this one and done. So this notebook also get created. Okay? Sorry, this pipeline also get created which is calling the customer address thing and let's click on debug this as well. So this will also take a few minutes to three four minutes and this will copy the entire.

50:09 Deepak G.: Data of the customer address CSV into our one bronze layer folder and that folder. We are saving the data in terms of a parque file format. Correct? So now If you want to see the advantage of the synapse, this is the advantage of the synapse. I am just reading the data with through a pipelines. I'm creating the notebooks as well. And everything is come on coming out to be very

smooth.

50:35 Deepak G.: Even this you know scheduling part is also coming very smooth because you have a activity directly notebook activity, which will just directly take the notebook, which is there within this synapse itself. So you don't have to do the mounting. You don't have to create the separate Link services for connecting to the data. So all that effort can be saved when you do it into the synapse World. Okay.

51:00 Deepak G.: So this is a very real world project example, which we are seeing right now for the synapse. Okay. So let it get execute. This will take a while. Once this will get execute it will be Our bronze layer get created now if we look at the diagram. We have done up to this point. Now. What I'm going to do. The next is we're going to write out one more notebook, which is trying to you know, transform something for the silver layer.

51:29 Deepak G.: So I will read the data again write a one notebook, which will read some data from this bronze layer. Remove some nulls remove some duplicates or you know, we do some kind of a filtration and then we save the data as a silver layer. Okay, and then lastly we create a one gold layer. We will combine the join the customer and customer address and do some kind of an aggregation and create our gold layer.

51:57 Deepak G.: Okay, so generally in the bronze layer you put the data as it is in the silver layer, you do a initial transformation removing nerves removing extra columns doing duplication filtering out some rows you do all that. And then in the gold Air you try to aggregate maybe put down a lot of joints doing the aggregation functions like Guru by all that and you create something which is needed by the our Downstream system, so here if we see we after that we're going to create a one more.

52:37 Deepak G.: Data, that is we call it like a gold layer. Our date and for this also again we use a notebook. And after that our gold layer did I get created will use the power bi so today we'll going to use the power bi as well. And we connect a power bi to this gold layer. Okay to create a flashy dashboard. Okay. So that's how and entire project we going to see through this. Okay.

53:08 Deepak G.: So let's have a one quick five minutes break because this is running in letter one quick five minutes break and then we'll come back. And

continue from here on so before going that let's just see that that this notebook also get executed if everything has been executed successfully if I go down to my developed tab, sorry. my data tab if I go down to this Landing tab Under The Landing if I go down to my project full location If I go to this bronze layer.

53:39 Deepak G.: Now I can see customer address data as well. I click on customer address and I can see a customer address parquet file format is also here. So you can see that so customer address also get the the parquet file format and bronze leg it created for the my customer and the customer address both got it. So this data is now available. Now next work would be is to create the silver layer out of this and then the gold layer out of this and then lastly we connect that with the power bi and we'll see one extra thing as well that how we can do it through some kind of a external table.

54:20 Deepak G.: So we'll create external table for bronze silver and gold Air as well and we'll see how that can work. Okay. So let's have one quick break of a five minutes and then we'll come back within a 5 minutes.

01:01:03 Deepak G.: I deleted should give me the option but it is not giving that's completely fine. I come down here. I click on this parquet file right click it and then I get an option load to data frame. I use this load to dataframe rather than choosing a specific file will give a folder only. Okay, so I remove this. part and I just say up to this point. Okay.

01:01:29 Deepak G.: So what I did is I say spark dot read dot load. And our bronze layer folder and from that I'm reading the customer folder and the format is parking. Let's just run this and see what it is coming in. This is what we are trying to debug rather than directly executing in first. Let's understand. What is the data is there then we apply some small transformation and then we see it into our Silver layer, so let's just see.

01:02:01 Deepak G.: We'll take a while to start the session and then execute. Again in this notebook the code will remain exactly the same the way you writing in the past part. So it's nothing new. You need to learn to write the code into the synapse notebook. So that is the same thing. So that's why we have a learning curve is quite less when we covered the ADF and the spark part.

01:02:35 Deepak G.: So synapse has a very small learning cow. Okay. It always when you start it start the session until almost take two three minutes on an

average. Let it take that time. Meanwhile, let's save this file notebook with the customer silver. Okay. So we remember that. Okay. This is a customer server. Notebook. Ideally, you don't need to write notebook again here, but just get the very initial state.

01:03:09 Deepak G.: So I just do it so that you don't have any confusion. Okay similarly on the similar lines. Let's create a one more notebook that is for the customer address because that is also will take some time to start. So I come down here go to this develop tab. And here you have a customer silver notebook right click on. Three Dots and click a clone. So the moment you say clone it will copy it one more time.

01:03:43 Deepak G.: And here we just change to customer address name here and remove that extra copy thing here. and the change which you need to be here is change the folder to customer. address Alright, so our four, I mean another notebook get great mean while coming to our customer notebook itself. Let's see. Nothing has been done. You can see that. It's almost 2 minutes. Our spark session is still getting started.

01:04:16 Deepak G.: But guys when you practice this in your account, remember the spark cluster, which you're going to use here is going to cost you. I think we saw Costa 100. 120 rupees per hour. So please be careful while using this so try to have your code ready everything then try to do it. So it will not cost you a big money. Okay, so this sounds to be ready and now it is starting to do the work. This is yet to create.

01:04:53 Deepak G.: Meanwhile, we have created a lot of things. Let me click on publish all. So that all the changes will get saved in between because we created multiple pipelines. You can see that we create multiple pipelines data sets notebooks. So let's say publish. So that everything will get saved. This is the original window size. Now you see. How bad this looks like if you have to do daily a programming on this or you know working on this sounds really tough anyways, so now you can see that it has read our parquet file correctly.

01:05:32 Deepak G.: In fact, I have not given the file name. I've given the folder name only customer only I'm not giving the dot Park is something. Okay. So this is a file name only sorry folder name only and in the bottom you can see that it is. done Created all that okay. awesome Now, let's just for a Time timing. We have to

do some kind of a small transformation. Let's just check. There are four some of the suffix.

01:06:03 Deepak G.: It is coming undefined and sum is coming Junior. So what I'll do is I'll just here say DF Dot. stink and then let's pass the distinct. Column name that is suffix. What it will do is if we execute this. This will display distinct distinct. is ideally DF thought testing takes

01:07:34 Deepak G.: Okay, so we can see that there are only these types of suffix. Are there unidentified undefined IV PhD second senior Junior. So let's remove all those rows where suffix are undefined. Let's display. select

01:08:44 Deepak G.: okay, so we got all the suffix all the rows were suffix are defined. So we have removed all the rows which is where the suffix for that person is not defined. Okay, so we got this data. Now what I'm trying to do is let's Considered as a one transformation and let's this is the clean beta because earlier it was a problem that every person has some you know, suffix is there.

01:09:09 Deepak G.: So if it is not defined that's a bad record and we've removed that bad record. So I call it data frame 1 is equal to DF dot where suffix is not equal to undefined. And then now let's just save this result. right dot save and give the path equal to this one Let's save this data into another layer or another folder which we call the silver layer.

01:09:50 Deepak G.: This is where my data get saved under that silver layer. Okay got the point. Now again, this has to done automatically every time so I'm not running this here. I want to call this thing from the my pipeline. So what I do is I come down to my filter tab. So integrate tab again. I create a one more pipeline I call this pipeline. as silver layer customer notebook, so I select here notebook. In this pipeline, I copy here. Go to the settings tab. And under the under the notebook.

01:10:47 Deepak G.: We select the notebook name as customers sell word notebook because that is what the silver layer code is here. We select the cluster spark cluster. Keep the executor size and that's it and let me execute this. So now this silver layer pipeline get execute the work is to read the data from the bronze layer remove all the records where the suffix is undefined. And then whatever be the clean data comes in save it into the cell World layer. Got it.

01:11:21 Deepak G.: So this will going to save my data into the silver layer. of the

one folder location Hmm now similarly, let's check for the customer address also. What is there in the customer address notebook build? Also customer for customer address also will just remove few things here and there and see what can be done. for transforming this customer address Again for this also it will take a little while to start the cluster.

01:11:59 Deepak G.: Let it take meanwhile our pipeline for customer layer is working. Let's just see where it went. Here it is. Still executing. Let's get execute. We have to wait till then. Okay available pool capacity is Okay, anyways, so what? Let the pipeline finish and then we retrigger this one again. Customer addressing it is saying that we have crossed the limit. of the number of sessions Let it get finished.

01:12:51 Dharmender S.: In the write command , we havent specified the file format . why it is like that ?

01:12:57 Deepak G.: Yes, that means we have not Define the file format because default format is a parquet file format. Okay. So if you don't Define the format by default, it's a parque file format. If you want to explicitly save the file either into the Delta format or into the CSV or a Json format then you could Define the format you say the data frame dot right dot format and then you define the format.

01:13:22 Deepak G.: If you don't want to give the format then by default it will take as a parque file format, which is a default format.

01:13:29 Dharmender S.: thank you

01:13:32 Deepak G.: In fact the same thing remains for the even for the read purpose as well. So even in the case of a reading also, it happens C. Okay, so if you don't Define any format for reading it consider as by default as a parquet file format. And we already seen why parque file format. We prefer because Parker file format is a columnar columnar type of data format where the data get saved in terms of a column wise and when you read the data or write the data in terms of of parquetful for me, it is faster and the space the parquet file format take is smaller as well.

01:14:16 Deepak G.: Still it is running running running running running running running too much. Yet not even started. What I'll do is I come down to this customer. let me just publish everything and then we stop the this park session

for this notebook. So that others can run. The publishing is still under the progress. Okay publishing it successful. I'll just say close it asking me to stop the session. So this session gets top So eventually now if you rerun this.

01:16:41 Deepak G.: Okay. white kid field Okay, there is some problem in the notebook. Let's just open the notebook and see what is the problem. Come down here onto the note developed tab under that there is a customer spark notebook open that. Let's see. Where is the problem? looks like I don't see any problem at such. Let me run it here itself. Oops, I should say cancel. Because this is a bronze layer thing, right? We have to run the silver layer.

01:16:53 Deepak G.: silver layer notebook Dataframe.bear suffix is not equal to this. This Define is fine the different the right dot. Okay, so here is the problem. The path should not be here. It should be something like this. Auth equal to so I just put the semic inverted commas before the path. So it should be after the path. That's why it was giving an error. Okay. So let's write a pipeline itself. Go back to our pipeline. Go to Silver earlier notebook. And click on debug again.

01:17:09 Deepak G.: So now this time ideally when it runs hopefully this will not create a problem.

01:17:47 Deepak G.: still running Let me get run. It's taking too much time. So meanwhile, it is running you can see that how many components we have created so far. We created the first pipeline. For moving the customer. Let us second pipeline for moving the customer address data, then we created a bronze layer. So this will call a notebook which is for customer bronze layer. We created one more pipeline that is bronzelayer for branch layer customer address.

01:18:33 Deepak G.: Now we created the silver layer customer notebook pipeline as well. Okay, if we talk about just only the notebooks thing so in the notebook you have a one notebook for your bronze layer customer one notebook for customer address bronze layer. One notebook for customer silver layer notebook and one notebook for customer address. Silver layer notebook, okay. So welcome creating multiple components. so far okay, so this pipeline still working for our customer silver layer Okay, finally this gets successful.

01:19:12 Deepak G.: So now this gets successful means is if I come down to my this data tab, if I go down to my Landing folder go down to the project for I should see silver layer in Project 4. I can see silverware if I go down into the silver layer. I

can see customer folder and there I can see a customer parquet file format.

That's okay. That's understood. That's not a problem. Correct. Great.

01:19:39 Deepak G.: now similar thing we have to do it for the customer address. So just quickly finish it for the customer address as well. So if you go down to the develop tab go to the customer address silver layer notebook. Let's just start the session here. This will take another sometime. It's too slow process. and once our customer address the silver layer gets Live session has failed session Illustrated error available your job requested 12 week course the pool on laser. I don't know. We already stopped everything.

01:20:26 Deepak G.: anyways Let's publish everything for now. So at least our changes get saved, then we'll refresh it. I think this error should gone because we are not running any other interactive session for this park cluster. So I delete this. Problems should not come. So let me refresh and ever sell up to Silver layer. We are good now. So if you look at this diagram. We have almost finished up to this point. Okay.

01:21:07 Deepak G.: let me just refresh everything Let's go to customer address. notebook This works too bad.

01:22:04 Deepak G.: Now we can see that on the top again. I am seeing them seeing the message. Please. Wait a few minutes. Session get failed. I don't know. What is the problem? Are using the same cluster it was working for other nodes. Is not working here.

01:22:45 Deepak G.: see this is working here. For everywhere. It is creating a problem. Levy sessions feel special available capacity your job request to 12 week or have the pool only has a vehicle out of the coat of the tool course. Try ending the running jobs in the pool reducing the number of week or requested. With jobs is running. That we are not very sure. Let's just count down here.

01:23:20 Deepak G.: see sparkles

01:23:51 Deepak G.: okay, you can see that this one is running. Somehow you can see that customer address notebook. This is running. You need to stop that somehow. This one has been keep on running. Which one it is? Say cancel. Yes. Okay. Let it get canceled. Now. Let's just quickly. Okay, this got canceled successfully now if you go down to over this code again, and if we try on our customer address silver Ideally should not give a problem.

01:24:47 Deepak G.: Ideally there was a one something was running that might seems out to be creating some problem. We close that out. Let's see. Now what we do is like once our the silver this kit done. We create a one small gold trying to combine all of these two things together and see if we can do some combination and then we create a gold layer and after that. We connect the power bi and we create the external table as well. Right now.

01:25:24 Deepak G.: We are doing everything through the spark way. Okay, you can see that this comes out to be ready and this is going to work now. Then we will see like how we can create the external tables as well. So we create that a data warehouse as well. So I'll just show you that. as well done so our this looks like to be done now good. Then start saying if one thing yep, what? Here you can see that in the customer address.

01:25:53 Deepak G.: We are getting something a data something like this. Let's just quickly see if anything specific address type has something. so I just said DF Dot select or testing likely address type see what are the different address types we have. Okay, we are two one is main office another one is shipping. So let's just take only the address type which is of a shipping type. So I just say era frame one is equal to data frame dot select.

01:26:54 Deepak G.: Let's type equals to main office Okay, so This will give me all the rules there. The address type is of a main office and then let's just lastly save this into over silver layer. So we did have one transformation where we remove all the addresses, which is not of a main office type and then we are saving this information into a path. That is our silver layer one. So I just copy this entire path. And I push it into a silver layer. I just say instead of bronze layer.

01:27:37 Deepak G.: I just remove the folder name here and I call it like silver layer within the silver layer inside the customer address folder. This should go. So again this get created perfectly fine. Now, what I'll do is I come down here create a one more pipeline go to integrate tab. Create a one more pipeline. Okay. We already have no the pipeline. We already don't have let's click on Pipeline. We call it.

01:28:07 Deepak G.: silver layer customer address pipeline and in this pipeline again, we use a notebook this notebook is selecting the customer cell no customer address cellular notebook. Say this spark pose spark cluster cluster

size. Took the minimum done. And then we did it. So now here we are calling a one notebook. That is a silver layer customer address notebook. And when I execute this this will going to Pull the data from my customer layer a customer address bronze layer.

01:28:49 Deepak G.: Do the transformation and push the data into the customer address silver layer? so this will see an our customer address silverware also get created after this. So let's just let it get execute and then side by side. I just come down here quickly. I just clone this customer address silver thing to make a one gold layer thing. Okay, so I call it. a gold layer So we created one gold and notebook. Now in this golden notebook.

01:29:31 Deepak G.: What what we can do is we read the data from both of them. So let's say I am reading the data from data frame 1. That is the customer address one. silver layer so here data frame 1 that is customer address and data Frame 2 is something which is reading the reader from the silver layer of the customer. Okay now both of them read the data and then finally I think we can do this. Join between these two.

01:30:28 Deepak G.: Where we can say data frame 1 dot join. data Frame 2

01:31:03 Deepak G.: What is that column name? Let's see that anybody remember that column name on which we can do this. I think customer address ID or what? Anybody following me if they can tell me what was that? CDF Dot Show for now.

01:31:27 Dharmender S.: you can print schema

01:31:54 Dharmender S.: df.printSchema

01:32:03 Deepak G.: This got failed. Let's just see. What is the reason? The one which we did got failed. Let's see. What is the reason behind that failure? The error which is showing that reason why it got failed unresolved column with the suggestion. A column or a function parameter with the name address type cannot be solved.

01:32:39 Srikanth R.: Name is differ

01:32:39 Deepak G.: Address type column was there. a column of function parameter of the address type name I'm sorry. I guess we have to use equal to

01:32:52 Srikanth R.: there is space

01:32:53 Deepak G.: let's just come down here onto our notebook itself and sort it out.

01:32:57 VEMULA M.: Main Office

01:32:57 Deepak G.: Go down to develop tab.

01:32:59 sanat s.: Need to apply where

01:33:04 Deepak G.: Go down to the customer address silverware notebook.

01:33:06 sanat s.: Rather than select

01:33:14 Deepak G.: Of hash I said df1 Dot.

01:33:15 Srikanth R.: Main Office

01:33:17 Deepak G.: so get to know that if there is a problem, we'll see here itself. Okay, so this is the problem. Let me just double click this.

01:33:34 urmi p.: put where

01:33:39 Deepak G.: one minute

01:33:41 urmi p.: not select for filter

01:33:42 Deepak G.: he have not so.

01:34:01 Niharika M.: Space

01:34:05 Deepak G.: Everyone. Got you. address type

01:34:13 Ritesh R.: There should be space Main Office

01:34:26 Dharmender S.: use the filter

01:34:27 Sudhakar D.: where needed there

01:34:34 Ritesh R.: And I think it should be in where filter

01:34:34 urmi p.: where

01:34:34 Dharmender S.: or where clause

01:34:40 Deepak G.: you are right. Thank you so much. Here we use a Veer class. Thank you so much. Okay, great. We got it. silly mistake Yeah, so we got this the data frame dot where address type is equal to main office and then done this is going to work right. Now. What I'm trying to do is I'm executing this here itself rather than running it through the pipeline. Okay, so that our silver layer will get create eventually.

01:35:14 Deepak G.: Okay, so this will create our silver layer. Awesome, the silver layer get create if you come down here on to the notebook. Sorry data tab go to Landing folder. Guru project folder for silver layer and under that silver layer. We got this customer address. Folder and there we got the customer address data got it. So now we quickly have our gold layer. So in the gold layer, we're trying to execute this information.

01:35:49 Deepak G.: So I just copy the rate of here and this session is already started in our this customer address. So I'll just use this for now to write the code and then we'll remove the code from here. Okay, so I'll write the code here itself that is loading the data from the silverware customer address and customer, okay. And let's just quickly see what is coming from customer customer ID.

01:36:17 Deepak G.: address ID Okay, I think so. We can combine base of the customer ID. I think so. Okay. For this customer because address and let's just see what is there in the customer table as well? Just run that too as well. So customer address and customer does not have any address ID column so we can join the data frame 1 but join era Frame 2 on In summary, tell me the syntax. I think I'm forgetting something.

01:37:06 Deepak G.: on equal to customer ID and I said be the frame 3 is equal to this. And then lastly I say data frame 3.

01:37:28 Venmathi R.: `df1.join(df2, df1.customerID == df2.CustomerID)`

01:37:28 Deepak G.: display 3 Let's Run. This will join these two tables. Yeah, so looks like joining is also happening properly. Okay, we got so many things, okay. so Let's keep this data only this joint data. And what I'm doing is I'm just saving this joint data as my gold layer. So what I'll say, I'll say that's graphene 3 dot spark Dot era frame Dot right dot save and then I save this information into our gold layer Okay, so we join these two things together right now. I'm saving it into my gold layer. Then the go layer.

01:38:21 Deepak G.: I don't need any folder, but I just say. use case 1 so this is my first use case. That's why. I'm doing this. Okay, so I'm extracting this code from here and put it into a different. Notebook, that is a golden notebook itself. This is my golden. Notebook get create now. I create a one more pipeline. I click on here. I create a one more pipeline. I use a synapse notebook here.

01:38:55 Deepak G.: This notebook is something which is calling my gold layer. gold layer use case 1 and under this gold layer notebook. We just select pipeline. We select the golden notebook. in the bottom golden notebook. We select the spark pool. It's like this. Now just because we are running little on short on the time I could execute this notebook. Sorry. This pipeline in this will create the gold layer folder and data inside but as we are short on the time, so I'm just running it from our this.

01:39:45 Deepak G.: Already existing folder here itself, which is already. This notebook which is already working fine. So I just run it from here itself so that it will create. Thus gold Air folder for me. What is the problem? Use case. Column already is the column modified it already exist. Okay. So it has creating a little bit problem at the time of saving saying that data frame one item to both have a column called.

01:40:21 Deepak G.: Modified on okay. So before that, let's just remove the modified. column from here So I just before joining this I just say I don't need that modified column for now. So I just say data frame 1 is equal to data frame 1 Dot. With column renamed and what was my column name? The column name was? Is having coming as a column has column name is. modify date I just rename this column from here and I say that is modified date.

01:41:09 Deepak G.: and that's Okay, and then let's do the join so that this will not come as a problem. Again a problem. Roguey ID is also a problem. so I just did for this Rogue we ID

01:41:59 Deepak G.: Okay, so you can see that now we rename these two column because these two column coming. So we just rename with modified data address GUI id address and then we join it on the customer ID column and we save the data into the gold layer. So if we come down here onto our goal layer. Landing folder project 4 I should see a gold layer I can see gold layer.

01:42:23 Deepak G.: And I can see the use case and I can see the data coming here. But it so we got all our silver layer gold layer and Brass layer. now let's publish all and let's just quickly one more time and then we'll see from the table angle. Okay, that is from the external table. So somebody has a data analyst who is working into the synapse how he can access it. Okay right.

01:42:49 Deepak G.: Now if you see the problem is we ever sell gold bronze layer silverware golden all the data is there by in terms of a files and for parque file format only. Okay, and we have a couple of Pipelines. Let's just bring everything together again here. Okay. So what we did we start from our CSV file? Our Azure SQL database will move the data from the database and we push it.

01:43:17 Deepak G.: for this customer data table you push it into a folder called Landing folder and that is customer folder. Then we created a second pipeline that moves the data for customer address from our SQL database. So you can

see that customer address data comes into the folder call Landing folder and that is a customer address folder now we created a notebook. Okay, so we go down to our developed and we call we Creator and a bronze layer notebook.

01:43:48 Deepak G.: Okay, what this notebook was doing this notebook was pulling the data from this Landing folder location Landing customer and push it into the bronze layer. Okay, so we created a bronze layer customer folder and we copy the data here. We did it for the address also. Similarly. So we move the rate of the customer address CSV into a bronze layer and that tune to the parque file format. Now who will call this notebook.

01:44:15 Deepak G.: So to calling this notebook we create the pipeline. We call it like a bronze layer customer address pipeline in this we created a one synapse activity that will call the notebook. Okay, which notebook customer address notebook. Similarly. We created a one more bronze layer pipeline that is for the customer address that is calling the customer notebook which copy the data from CSV to the park a into the bronze layer.

01:44:42 Deepak G.: then we created the silver layer. Now what we wanted we wanted to do some kind of a transformation. So we created a one notebook called customer silver notebook in this notebook. What we try to do is we try to remove all the rows there our suffix are undefined. So we need so first we read the data from bronze layer. we remove all the unwanted items like where the suffix are undefined and then whatever be the result comes in. We try to save these result into the silver layer of the customer got it.

01:45:18 Deepak G.: So we save it like silver layer customer folder on the similar lines. We created a One customer address silver notebook what we did there in the customer address silver notebook. We again read the data from bronze layer customer address. We find out all the rows where address types are main office only and then we save it into the silver layer of the customer address. Below data is just for the testing purpose.

01:45:46 Deepak G.: So let's just remove this that is not needed. Okay, now finally to call this Silver Lane notebooks as well. We created the pipeline. So one pipeline is silver layer customer address that again call the Customer address notebook then we have US Cellular customer pipeline for customer, which is calling the customer notebook. Then lastly we go to the notebook and we create a gold layer

notebook. So in this gold layer notebook.

01:46:16 Deepak G.: What we did we read the data from the silver layer of customer address. We read the data from Silver layer customer folder. We did the join on these of two things. and then we try to save the result in terms of the gold layer got the point. So this is how we save it into the gold Air and calling for this notebook. Also, we create a one pipeline you call it gold layer Pipeline and the gold layer pipeline. We call this gold layer notebook.

01:46:46 Deepak G.: So end to end our everything get create now. Our data is into the gold layer. before connecting it to the power bi let's just see from the this snaps angle of the external table angle. So if a data analyst comes now think from the data analyst angle data analyst comes and you want to play around with the bronze layer silver layer and golden into the SQL terms.

01:47:13 Deepak G.: Can he do it right now? He cannot do it because if you see in the workspace, you got a once equal database. That is our a sequel because of ours as serverless SQL pool and in that database we can only have the external tables. So let's let's create a One external table. Okay, so let's go down here and let's create this time instead of anything else. Let's create a sequel script. Okay.

01:47:43 Deepak G.: So we're going to create a once equals script what this equals script will going to do is let's create a three schema. Let's say we want to play with the SQL as well. So what I'll do is It got stuck. We'll create a schemas. I don't know why it is not expanding. Okay. Yeah, it is expanding now. got stuck Okay. We create a three schema. create Sema that is prawns and the note run this a bronze get created Okay, so in the bottom you can see here.

01:48:51 Deepak G.: When I run this bronze schema, it has created a one bronze schema. Let me create similarly a silver schema. Echo this will get the silver schema. And let me create a one more schema. I call it like a goal

01:49:11 Deepak G.: cut it. So now if I look at the database side also. This is my SQL database. If I refresh this. I go to external. Sorry I if I go to the schema in the bottom. It's not yet refresh. Sorry, I'm not selected the correct database. So on the top, let me select the bat 7 database which we have let me now quickly create again. bronze to create the bronze Let me create the silver.

01:50:08 Deepak G.: sticks too much time You can see that that's still. Executing. Okay now I'd get great. Let me click again cellular layer. So I'm creating a three

different schema one for bronze layer one for silver layer one for gold Air. We did Via parquet. We I mean our spark way. Let's say if I want to do it through a data analyst way if I want to X It's all of that how I can do that. So I created three schema here. I refresh this.

01:50:46 Deepak G.: I can see bronze silver gold and on the top I can see bronze layer. Okay, we got this now you see a one Beauty here. Okay, so that is something which I wanted to show you. For now I call this. Publish all for now. Let's publish all. Because we are doing a lot of changes. So let us get saved. So it won't be a problem. Okay. Everything got saved. Okay, this will get saved.

01:51:28 Deepak G.: now what I can do is I can basically Go to a specific folder location. stabilizations I come down to this data tab. I can go to the link. I go to The Landing folder. I can create the external tables for accessing purpose for the bronze layer silverware and everything. I go to Project four I go to the bronze layer. I go to the customer folder location. I right click on this and I can say here.

01:52:06 Deepak G.: You can see that on the customer folder of the bronze layer. I right click I say create external table. Now what it will going to do is it will create an external table as well on top of my bronze layer. Okay. So rather than only existing the bronze layer via spark way I can access it via SQL weighs will so I just say here give the name as bronze Dot.

01:52:31 Deepak G.: customer open the script you can see that it is creating the external table again. I'm not going into this too much because yesterday we already seen that it first is create data source. Second is create the format. And you have to just execute it. So automatically it will create a table for you. That is a bronze layer table and you see the data, okay. now the same thing we're going to do it for. Our second table as well.

01:53:01 Deepak G.: Customer address right? Click create the external table And this is a bronze layer for the customer address. I say continue. I see here. I say bronze Dot. customer address then I say open the script executed. So this will going to Do it for the customer address? So you can see customer address table also get create. Now come down to that folder location Landing folder now similarly do it for this silver layer. Here go down to the silver layer.

01:53:41 Deepak G.: Go down to the customer folder right click. SQL script create external table now this time we give the name as silver Dot customer. So this will

create this silver layer customer table. say execute so silver layer customer table also get create now again, go back to the folder this time do it for the silver layer customer address. So come down SQL script create external table. continue silver Dot customer address open script run and finally, I go down here and do it for the gold as well. So project 4. Come down to the gold layer.

01:54:44 Deepak G.: use case 1 create external table this time. I give the name as a customer full let's say so I just say hold on. customer full Open the script. and execute done got the point now here you see the beauty so we created this multiple tables now if I go down to this workspace if I just refresh this workspace. See the external tables. You can see I start seeing all the tables bronze customer bronze customer address silver customer address. If I right click on this.

01:55:38 Deepak G.: I see select top 100 rows of this Branch table. And I execute I can see the this data is coming from my customer address folder tape. as customer address sorry this coming from customer folder from the bronze layer. You can see that so the data analyst can also be visualize my data or see the data completely by doing this external table. Initially, we created the load the data by ADF.

01:56:09 Deepak G.: Then we move the bronze layers silverware gold layer. We're doing all together into the spark world. But side by side what data unless can do is the if he knows the folder of the bronze layer silver layer gold layer. He can create an external layer on top of it and he can you know, see what is there in the bronze silver and gold Air for all these specific type of tables so you can see that this is so easy now, so if I can just want to see the customer address information for the bronze layer, I can just say simply execute.

01:56:40 Deepak G.: And this will going to work for customer address got the point. So the same thing can be usually visualize here from this SQL perspective as well. Whatever the folders which we have created using this part. Got it. Now. Finally, let's connect this whatever the silver layer bronze like gold a Vivek created through a power bi. Okay. So for the power BF what we need we need the power bi so what you can do is you can download.

01:57:09 Deepak G.: our bi comes in a two version one is a workspace a online version of the what is a desktop. I think desktop is free so you can download the power bi desktop for free if you want to make a power bi. The web account in that

case you need to have the it does not use a Gmail ID but minimum asking you have some domain name, but Power bi desktop is free.

01:57:35 Deepak G.: You can download it and you can use it. So this is what I did so I have power bi install on my machine. We just download it. This was the URL. Here you click on download Power bi for desktop. Click next power bi Again download. and then Just open the power. We have desktop. I open it here. So let me just quickly open my power bi again.

01:57:59 Deepak G.: I am not at all someone who is use the power bi a lot. I have just in a very high level I've used it because I'm not a data analyst so sorry if I do some mistake in the power here and there. Okay, so I'm not a power bi Pro. Okay. So, let's see how we can kind of power bi with the our synapse same power bi can connect with the databricks also. Okay.

01:58:24 Deepak G.: So bobier is starting up hopefully yeah, so it is starting up. I guess multiple time got clicked.

01:59:02 Deepak G.: Remember one thing in our synapse. We have this a serverless SQL pool. You can see that we are connected to the silver serverless SQL pool. Okay. That is what we're going to connect so I click on the manage tab here. Go down to the SQL pool. Under that SQL pool. I see this built in SQL pool again repeat manage tab. So this is my managed tab SQL pools in this built-in SQL pool. Click on this built-in SQL pool.

01:59:33 Deepak G.: Once you click on here here you see the properties of the SQL pool. So in the properties, we need to have the name of that SQL pool. Okay, so this is my SQL pool name. I can copy the SQL pool name. And here you can see that admin login is there okay. So either you can use the same username password, which you have given at the time of creation of the synapse account or you can use it by your ear user ID itself. So, let's see here. This is my power bi screen. Okay on the power bi screen. I can go to this top here. Get data.

02:00:18 Deepak G.: so I click on get data. Under that I click on the more. Now when I click on the more button. I should see multiple data sources. Out of the data source. We have to select the Azure one. So there you see a lot of azure related data sources.

02:01:00 Deepak G.: click on data click on more seems out to be hanged little bit. Okay, you can see that the window comes up when I click on get data more here

on the bottom here. You see azure. So I select this azure. And Azure here. You see synapse SQL analytical analytic sequel here on the second number Azure synapse analytical SQL. So let's say for example instead of this. I want to connect to the Azure data bricks.

02:01:53 Deepak G.: I want to pull the data which is there in my Azure Delta Lake if I want to access that I can use the Azure database and then I can connect to Azure width I connect and I can pull the data from the Azure databricks into the power bi. But let's say for now. I want to connect to the synapse only. So here you can see there are multiple option. Your database is blob storage. You can pull data from multiple ways.

02:02:18 Deepak G.: So let's I select the synapse one because right now we are concerned with selects click on connect. Okay, now it will ask me this sequel server database name. So which SQL Server database you want to? Get in so we go down to our eyes as I said we go down to our man monitor tab. Sorry manage tab under that manage we go down to the SQL pool.

02:02:48 Deepak G.: We click on the SQL pool that is built in one and I can see here the name of that sequel endpoint copy this endpoint name and That endpoint name here into the SQL Server database. Okay, keep the database option. Empty here. It's like trying to check. What is the connectivity mode? It will do a direct query like it directly fire a query to the build to the your synapse built-in database and get the data or it will import everything here.

02:03:20 Deepak G.: So let's do a direct query for now because we don't want to input the import the entire database. So we choose direct query option here and let's say okay. Now this will ask me how you going to connect e so the best option either you can use the database username password or else you can connect via Microsoft account. So I choose a Microsoft account because I will login with here with the same account with which my Azure account is there. Okay, so I click on sign in.

02:03:56 Deepak G.: So I this is Deepak will 11 at Hotmail. This is through which my Azure portal account is there so I choose the same account. As I already login into my system, so it automatically takes my login. And it won't ask me the username and password because I already login onto this machine and this will allow me. So now once this option get comes in you you are currently signing is

coming and the button is enabled connect. I click on this correct button. If everything goes fine.

02:04:28 Deepak G.: I will make a connection to my Azure synapse. Okay, and then I can pull the data from the Azure synapse and create my amazing dashboards here. Okay, so it's taking a little while to make a connection. Now you can see that we got this Mission 180 batch and snaps. Connect this is my database name. I can expand this you should see all the tables and schemas and everything.

02:04:55 Deepak G.: So here you can see that I am able to see my bronze customer bronze customer address a silver customer silver customer address gold layer. I collect on this data. It's trying to do the preview. If everything goes fine. We should be able to see the data. So this is the data coming in. We just say load the data So eventually our data get to it up now once the data is loaded up.

02:05:22 Deepak G.: We can start using a creation of this. This dashboards thing so we can choose anything chart of our own and we choose the rate out of it and we create our dashboards out of it. Okay, so let it get loaded up everything. So our customer data is coming here on the left hand on this here if you can see the screen. So here my entire data is showing it up. Okay.

02:05:50 Deepak G.: So what I'll do is let's say I want to have this chart. So I create I expand this chart in the bottom. I can fill in excesses and y axis. so let's say I don't know anything big to do it here to be honest. but let's say just to a customer address want to access so let's put it on the y-axis customer IDs count of this and let's have the middle name in the x-axis Okay, so Again, the information is too real irrelevant.

02:06:44 Deepak G.: But yeah, let me just think of something good. So let's say have another chart so that is a pie chart. Let me add a one pie chart here. So in this pie chart, let's say I will add the values from this suffix because we have a limited suffix, right? So add that suffix values here. You can see that so if I just expand this somehow. You can see that the suffix is coming properly here.

02:07:20 Deepak G.: Okay, so Junior majority of the rules are Junior. Then the second category is how much seventy twenty one percent Exacta Etc. Got it. So that's how you can keep on creating your charts. Okay. Now the beauty here is our data is staying where our data is in the synapse. Now, this is how the whole

integration can be done. We pull the data from on-prem to the cloud. We create bronzeless gold layer.

02:07:50 Deepak G.: We clean the data everything make it you know, a downstream business really now this a data analyst people comes in who is the power bi guys he would connect to our synapse thing or our data mixing it can connect to anything anyways, so they connect to our gold layer data and they start fetching the golden data. And show it in form of a some very good dashboards and this entire solution can be given by this.

02:08:18 Deepak G.: So that's how you can connect and to end and that's how a real world project works. Although this power bi part is not come to our bucket as like a data Engineers this something our work remains till the either the synapse or either the databricks side. Okay, after that connecting to the databricks or connecting to the synapse would be easy thing and that data analyst people will do it on their own by this way. Similarly. This is the example today.

02:08:44 Deepak G.: We seen it from the synapse world the same thing can be done for their project three, which you have it right in your in your account the project you are Delta Data lake house architecture in that also you created this gold layer right in that case. If you want to connect to that gold layered the database of your database that can also be done. So in that case, what you need to do is you go to the data here connect to that as your database.

02:09:12 Deepak G.: So right now we are connecting to the Azure synapse account there here you see the connector for the Azure database as well and the exactly the similar way you can connect to the Azure database also and then this will work end to end for from The Edge side as well. So I go to the Azure here. I see as your data breaks say Digital database and say connect and then you give the databricks up workspace endpoint the credential and then it is going to work it there as well.

02:09:42 Deepak G.: So here you can see that you need to have that database information here. And once you get the database information and you make a correct, this will going to connect to the databricks world as well. Okay, so there is nothing big as an exercise. What you can do is whatever the the project through you implement on that on top of it. You can connect the power bi through this Azure database way. Okay.

02:10:07 Deepak G.: So that's how this end-to-end thing can work. I'll just quickly recap what we did and then we get into the Q&A. Okay. I hope this was a little insightful you got something new thing today, okay. Close, so, let's see and this was our entire architecture. We pull the data from database. Using this synapse ADF pipeline push the return to an ad dealers account and then we create a notebook that notebook read the data from landing and push the data into the bronze layer.

02:10:41 Deepak G.: So we here we have a CS filter then we write it as a parking the bronze layer then we write the notebooks and then notebook is calling from the pipelines. Same thing. We did it from bronze later to the silver layer again. We write a notebook. We remove we do some kind of a transformation move the data from the bronzer to the silver layer. Why spark only then again we One more notebook that is combining the customer and customer customer address you both of them do the join in the spark itself and push the data into a gold layer.

02:11:14 Deepak G.: Now, you did everything to the notebook, but notebook every time rather than calling the notebook directly because you cannot schedule The Notebook we call this notebook through the synapse pipeline. So in the synapse pipeline, we add that notebook and we are executing the notebook through the synapse Pipeline and then you can schedule it. Also if you need it once we created all this Then what we did for we created the external tables as well. So we created a three schemas.

02:11:39 Deepak G.: So if you go down here, if you go down to our data tab we go down to our workspace here in this database. We created the three schema one is bronze. Saying great schema bronze create schema, silver create schema gold because the three layers and then we created the for a bronze layer for customer. We create the bronze customer table bronze customer address table similarly silver customer table silver customer address table and then for the gold layer we created the Gold Dot customer Full Table.

02:12:11 Deepak G.: Okay, how we did it we go down to that folder locations. So this is my landing folder. When we go down to the landing folder location where to just say, let's say this is my folder location on top of folder location. You just say create external table you get an option and you there you give the correct name? Okay. That's what we did. Okay. So for example, we were onto the project

4 we were to create a external table for customer for the silver layer. So I right click here customer.

02:12:46 Deepak G.: I say create external table and I say silver layer dot our silver Dot customer. So it will go into the silver schema and the customer would be the table lamp. That's how we created all those table once all those stable get created. We download the our power bi desktop and the power B8 stop we go to the get data on the red we go to the more options under that we select the Azure and within that Azure, we choose the option of azure synapse.

02:13:14 Deepak G.: Okay, and when you choose the Azure synapse option, it will ask you the Azure synapse and point how you get the endpoint you come down here on to the manage tab. Under the manage tab you click on SQL pools and you got your built-in SQL pool. Click on the built-in SQL pool and within the built-in SQL pool here. You see the workspace endpoints. So you copy this endpoints go to the here select Azure synapse.

02:13:41 Deepak G.: Say connect and then whatever be it is asking to connect there. You just put the synapse. Server name? Okay. So this this is how you connect and say okay and they will ask you the credential you choose the Microsoft account and whatever be the current account you just connect through that. So automatically you will have an access to see all that data. And once the data is there you select which table you want to see come down select that table select the chart which you want to print or you know create and that and then your power will be I think get connected. Thank you so much. I know we have extended little bit.

02:14:18 Deepak G.: But this something we need to finish up. Okay for the next week onwards. Will still have a session but next week my new batch is getting started. So we probably will not able to do it on the same time. We push the I feel at least two weeks thing left maximum so we cover these two week thing on to Coming week in the next week, but we're doing the sessions into the evening time.

02:14:44 Deepak G.: I will share that evening time. Okay, and if I try to arrange if my partners Arab is available in the morning, I'll ask him to take the fabric sessions so that it can cover in the morning itself so that you don't have to you know change the time and then the system analysis and is so we have to cover

the fabric. We have to cover the system analysis in design and we have to cover the chat GPT and ILM bar. So these three things are on my cards. So hopefully we're going to finish.

02:15:10 Deepak G.: That into our next four sessions if we need to feel that we need to extend a little more. We will extend it for one more week as well. So that's completely fine. Your course will get finished as we promise. And your what's a group is still open. You can ask all the questions and we'll still go in and be in touch. Okay, so nothing to worry and please if you like this session like in the efforts which we are putting in and if you feel that this session is helpful, you can share this with your friends and colleagues so that they can also be a part of this amazing course and you know can learn and make a shift in their career. Thank you so much.

02:15:49 Deepak G.: And when you do refer you get the referral cash bonus as well, okay? so let's get into the Q&A and what do you guys can do is like? Just up let me know in doing that now week time weekdays as well. If you are facing any challenges into the interview or something and if you want to connect for a one more bonus session during the weekdays, we are open you just tell us if you have a 1500 people. It's ready to come.

02:16:14 Deepak G.: We'll create a one more bonus session within the week itself in the evening time. Okay. I'll be available. Okay? Let's get into the Q&A. Thank you so much. Okay, let me start. Okay, I think I'm I miss one more thing. So I think I should show you the get devops get here as well. But time was the problem anyways, so if we can connect this to the devil's good everything can be saved there as well. Okay. Let's just get in the Q&A. So your mics are open guys.

02:17:01 Deepak G.: Go ahead and you can start sharing a screen if you can share your videos. You can switch on your videos as well. We can see each other also. It's almost our normal time. This is a one last session and then we get into. The different timings if you if you are fine, we can you can open your cameras as well. Okay. Yeah Ritesh's triveni, please go ahead and even a few.

02:17:27 Deepak G.:

02:17:28 Ritesh R.: Go a high Deepak. My question is instead of using a SQL pool. We can use Azure SQL and create the TA table and all this there. So the the charges will not apply.

02:17:42 Deepak G.: I just people also has a charge right? I just equals not free.

02:18:25 Deepak G.: into Ada. Let's location and we just pointing to that location for the reading purpose only. We're not making that we are not owning that thing. On the fly. So you are not owning the data you are running it query on the fly. Like you're creating here in the this We have not seen the dedicated SQL built-in sequel pool. pool. Okay. Okay. external table where data is residing create the tables of your own.

02:18:33 Deepak G.: So in the radicates equal you can So right now we are creating only the Okay in the external table.

02:18:37 Ritesh R.: And the other question is the in databricks. We are using our Delta table, right? So only in database only we can we can create the data table and here we can create the external table.

02:18:50 Deepak G.: Now here also you can create the Delta table. Delta table is a format understand one thing. External table or a manage table is a two types of the table. I can have the you know external table which is Delta format is the way the data gets saved. So see your data can save in this Json format. It can save into the parquet format. It can save it into the your CSU format and it can have a Delta format as well.

02:19:22 Deepak G.: You think like that but the Delta will give you some extra capability also, so even you can create an external table on the Delta Lake format as well. There is fairly possible. So for example when I was creating the tables here Let me just quickly show you. So let's say I go down to the SQL script what I was saying here. create this External file format right now format type is parque, but this format type if you are it has even the Delta format. So then format type could be here. You'd say it like a Delta.

02:19:59 Deepak G.: Okay, then your extract external table get create which understand that underlying the days save in Delta format. So when I read the data, I should really using the Delta format. Okay.

02:20:11 Ritesh R.: Yeah, thank you. Book. Thank you.

02:20:13 Deepak G.: Okay. Any more questions anyone?

02:20:23 Triveni B.: Yeah.

02:20:23 Deepak G.: through in

02:20:23 Triveni B.: Hi Deepak. I am triveni. So Deepak, could you please share

the recording sessions the link the Deepak from the long back I missed you were training recording sessions. So I forgot that site Deepak.

02:20:38 Deepak G.: what is your email ID?

02:20:41 Triveni B.: beat revenu 574 Deepak

02:20:44 Deepak G.: one

02:20:45 Triveni B.: Yeah.

02:20:49 Deepak G.: you 574 Yeah, I'll just send you an email.

02:21:01 Triveni B.: Sure. Thank you Deepak.

02:21:03 Deepak G.: Thank you so much. Yeah, any other question anyone?

02:21:13 Hemant R.: But hemant here, so this one request if you can last take the data architecture overview session.

02:21:23 Deepak G.: Data architecture overview session. I have not getting it what exactly you want to know there.

02:21:28 Hemant R.: I think in one of the meeting we discussed because during the higher experience some data architect related questions.

02:21:35 Deepak G.: Okay thought

02:21:37 Hemant R.: They are asking so you sold at my separate courses there but I will just happen our session on that one how to prepare that one like what process? Yeah.

02:22:02 Deepak G.: mrttable guys

02:22:04 Hemant R.: yes, if

02:22:05 Deepak G.: Yeah, please.

02:22:08 Hemant R.: that's it from my

02:22:09 Deepak G.: Okay. Anything else anyway?

02:22:15 Pati M.: Here when we are writing data. It is saved in the format of Snappy dot pocket. So what is the difference between Park and snappy.com? Is there any

02:22:25 Deepak G.: snapping the operation format That's a compression technique it's used. The format is still remain the same. Okay.

02:23:23 Deepak G.: Any more questions anyone?

02:23:26 vivekanand d.: Deepak

02:23:27 Deepak G.: Yes.

02:23:29 vivekanand d.: suppose say a project is to be like newly Incorporated.

So like can we do it?

02:23:37 Deepak G.: Sorry Come Again Come Again.

02:23:40 vivekanand d.: Suppose say we are doing a proof of concept for a new or RFA. Okay for a project say suppose Healthcare domain. Oh, can we do the and tired like into databricks itself without any of

02:24:02 Deepak G.: So that depends upon what work you are trying to do.

02:24:08 vivekanand d.: so like how to understand that like

02:24:11 Deepak G.: A logic is simple whether it is a POC or it is not a posit doesn't matter. Okay, it's important that what you're trying to do.

02:24:18 Deepak G.: Let's say you are trying to pull the data from a variety of data sources and then you are trying to write the transformation logic another another logic or it in that case pulling of those data from those variety of data sources need to be done through an ADF. Okay, and the transformation which you are doing or the Logics validations all the aggregation what you are trying to do could be done into the databricks.

02:24:41 Deepak G.: Okay, just what UC purpose you assume that okay. My data will automatically come from the let's say Salesforce or a rest API or something. So I expect that whenever my data come from that this will be some exposite file will be there. So for a timing you can put that x y z file into a one folder location and assume that that file come and then you can do everything into the databricks for just for the POC purpose. Okay.

02:25:05 vivekanand d.: Okay. But so which means like we cannot sideline ADF Etc because there is a capability of data.

02:25:16 Deepak G.: Again, it depends. What is your use case? Okay. ADF is more of an ETL. Okay. So if you feel that your project is something which not need any ETL thing then you can completely skip the ADF but if you feel that the ADF portion is there in your project then you it's preferable to use an ADF. There will be not.

02:25:36 vivekanand d.: No, no.

02:25:37 Deepak G.: Oh Smart in doing the lot of ETL okay it

02:25:43 vivekanand d.: That is not what I meant. Like it databricks also has some inability ETL capabilities, right?

02:25:50 Deepak G.: It has it, but it is not as strong as ADF.

02:25:54 vivekanand d.: Oh, okay. That's why we are always like connecting it with area.

02:26:01 Deepak G.: exactly Okay, any more questions anyone?

02:26:11 Gopinath M.A.: Deepak assignment solution 7 is missing in the portal.

02:26:15 Deepak G.: Oh, yeah. Spark streaming. I did it yesterday you got that. I just add this as well. sessions and we haven't anything else is spending from Iron guys is let me know.

02:26:33 Gopinath M.A.: Thank you.

02:26:36 Deepak G.: Thank you. Okay. Any more questions anyone please? Yep. Okay, if no questions left we can call it a day. Thank you everyone. Thank you for joining in.

02:27:04 Srikanth R.: Skip and get the book for the great

02:27:04 Naveen T.: of time

02:27:06 Deepak G.: Thank you.

02:27:07 Srikanth R.: session. Thank you so much.

02:27:08 Deepak G.: Thank you so much. Thanks. Bye.

02:27:10 Vaibhav A.: Anchorage book

02:27:10 Deepak G.: Thanks. Bye. Thank you.

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